Auditing Government Ad Delivery Skew in Practice

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In recent years researchers have been concerned with skew in ad delivery, whereby ads are disproportionately delivered to particular subgroups of users over others, despite advertiser intent to reach user subgroups proportionally. Online ad delivery platforms are not expected to show ads to user subgroups at a proportional rate; indeed, advertiser and consumer utility may benefit from ad delivery that is powered by algorithms that predict ad relevance. However, when ads contain consequential information such as opportunities for housing and employment, skewed ad delivery can exacerbate pre-existing societal disparities or amount to discrimination [1]. In addition, while not legally regulated, skew in ad delivery might be detrimental to the mission of government advertisers that use online advertising to raise awareness about public services. In our work, we present a case study and practical framework for government advertisers to audit their advertising campaigns for skew in ad delivery. Government advertising is a high-stakes setting, as access to information about public goods can materially improve social disparities. However, the domain has been understudied relative to contexts such as housing and employment since government communication is unregulated.

To address skewed ad delivery, researchers have developed methods to audit ad delivery through field tests [4, 5]. However, these methods were developed without considering the cost of audit field tests and the organizational challenges that an advertiser might face in implementing the field tests. For example, while existing ad delivery audits use field tests that target users with known demographic identities, some platforms, such as Google Ads, label a substantial share of users as "Unknown." Users are assigned the unknown label if the platform has not identified them as belonging to a specific demographic group. In some cases Google relies on users' self-selected settings, while in other cases Google infers demographic labels based on browsing behavior [3]. The accuracy with which Google assigns demographic labels is uncertain: in one 2018 survey, researchers found that Google Ads male and female labels were often accurate [7]. However, researchers have found that Facebook inferences of user interests and data broker sourced demographic labels are often inaccurate [6, 8]. As such, past work has proposed ad audit methodologies that exclude unknown users and rely on alternative baselines for user identities, such as through voter records [4]. However, voter records are not publicly available in many states; instead, advertisers often use platform-supplied demographics for targeting. Because most advertisers are incentivized, but not compelled, to adopt audit field test methodology, it is important to consider if audit methods realistically suit advertiser budget and strategy. As such, it is important to consider how advertisers can best audit and address undesirable skew in ad delivery in the context of platform-provided user demographics.

Our Contributions Our contributions are two-fold. First, we propose an auditing framework for government advertisers that is responsive to the organizational practices and competing real-world priorities of government advertisers, with particular attention to the "unknown" users on the platform. Second, we conduct a case study in partnership with a state-level government agency, in which we split a campaign into sub-campaigns to measure and reduce skew in ad delivery, while also considering cost trade-offs. Our approach strategically targets unknown users, achieving meaningful reductions in ad delivery skew while reducing costs. This approach helps mitigate the trade-off advertisers face between cost efficiency and control in ad delivery, enabling advertisers to address ad skew with a reduced cost premium.

Methods We partnered with a state-level government agency to measure and address skew in ad delivery of an advertising initiative for small business owners. The initiative advertised a government website that contained information about taxes, supportive services, and grant opportunities for entrepreneurs or people planning to start a business. Historically, women and racial minorities were underrepresented among business owners in the state. The government agency sought to prevent worsening disparities in business ownership by ensuring their ad initiative reached users at a rate proportional to state population demographics. Our research question asked: can a campaign budget split enforce parity in ad delivery between male and female users without increasing advertiser costs? In our case study we focus on gender on Google Search, where users can take one of three labels: Male, Female, or Unknown. Recently, budget splitting has been proposed

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as a cost effective and interpretable approach to reducing skew in ad delivery that is accessible to advertisers with low technical proficiency [4]. Our approach to budget splitting follows the following steps: (1) achieve organizational alignment, (2) measure skew, (3) define budget split arms.

Achieve organizational alignment Government advertisers often outsource their advertising campaigns to vendors. While government teams set priorities, their vendors have more direct control over the implementation of the campaigns. As such, during vendor procurement, government advertisers need to ensure the partnership will enable future testing. This includes setting expectations for data collection, access and format; reporting cadence and availability; and campaign documentation [2].

In our case study, we achieve organizational alignment by mediating conversations between the government agency and their vendor. Our case study also demonstrated several concerns that are common to government advertisers: vendor management and low in-house technical capacity. In particular, we specified the structure of the data and granularity of engagement (by gender group) and the campaign documentation that was necessary to measure skew.

Measure Skew To assess skew in ad delivery, we draw from the delivery ratio used by the Meta Variance Reduction System [4]. We generalize the delivery ratio definition to an engagement ratio, as advertisers may seek to enforce parity at alternative stages of the advertising funnel (i.e., clicks or conversions), whereas the delivery ratio only pertains to impressions. The Engagement Ratio (ER) for a given group g and advertising campaign c is defined as $\mathbf{ER}_{g,c} = \frac{\text{Engagement}_{g,c}}{\sum_{i:x_i=g} \text{Engagement}_{g,c}}$. $\text{Engagement}_{g,c}$ corresponds to the engagement (e.g., clicks) attributed to group g for advertising campaign c.

In our case study, because the government advertiser is interested in equal opportunity to the supportive business services, we focus on parity at the delivery level (impressions). As such, the engagement ratio serves as a delivery ratio. In particular, we report the delivery ratio of male users, relative to all impressions by male and female users. We exclude unknown users from the calculation of the delivery ratio because we seek to approximate the distance from our baseline, state demographics, for which there is no parallel to an Unknown user. We thereby consider ad delivery to reach parity if the 99% confidence interval for the male delivery ratio includes 0.5. If advertisers use alternative baselines, such as the distribution of users in the eligible audience, the above definition could be expanded to include Unknown users.

After cold start/learning period, use evaluation metrics to redistribute budget across split arms.

Define Budget Split Arms We split the campaign into multiple sub-campaigns to control budget by demographic group. Since some advertising platforms, such as Google Search, do not enable advertisers to target unknown users as a standalone group, we propose splitting a single campaign into a set of campaigns that each target either a single known demographic group, or a single known demographic group coupled with unknown users. For example, in the case of gender we split a campaign into four sub-campaigns with different target audiences: male, female, male and unknown users, female and unknown users. We avoid increasing competition by iterating through two cycles: cycle A with two active campaigns, one targeting male users and one targeting female and unknown users; and cycle B with two active campaigns: one targeting female users and one targeting male and unknown users. We alternate the day of the week and time of day for which each cycle is active. The budget for each sub-campaign can be calculated with the desired baseline ratio and subgroup cost effectiveness.

In our case study, the desired delivery ratio to male users is 0.5, to maintain consistency with local demographics. In the absence of priors, we initialized campaign budget with spend that allowed for 50% of the budget to be attributed to male-targeting campaigns and 50% of the budget attributed to female-targeting campaigns. The budget allocated to each campaign should be iteratively updated as the advertiser learns the cost effectiveness and skew present under budget splitting. The budget should then be updated according to group cost effectiveness: for example, if female users have a higher cost, then they will need to receive a greater share of the budget to achieve the desired delivery ratio.

The vendor-set marketing strategy aimed to drive volume of clicks and conversions through two campaigns: one with a max clicks bidding strategy, and one with a max conversions bidding strategy. We will refer to the campaign that maximized clicks as "Max Clicks" and the campaign that maximized conversions as "Max Conversions". We split the budget for the two initial campaigns into eight sub-campaigns using the methodology described above. Furthermore, to schedule alternating time slots, we separated each sub-campaign into two campaigns, which were identical except for schedule (cycle A, cycle B). We leverage our

framework for budget splitting to ensure that at any point in time, users (male, female, or unknown) would be targeted by two active campaigns (Max Clicks and Max Conversions), which was consistent with the targeting setup prior to budget splitting.

Empirical Results We measured skew in the original campaigns (having no budget split) over 31 weeks, and identified that in aggregate, ad delivery for the Max Clicks campaign systematically skewed toward male users. Figure 1 shows consistent skew (y-axis) toward male users over the 31 weeks (x-axis) for which the original Max Clicks campaign ran. The original Max Conversions campaign, however, did not demonstrate consistent skew. Our results confirm prior work demonstrating persistent skew in ad delivery despite advertiser intent to reach subgroups proportionally, and that skew may be shaped by facially neutral ad strategies like bidding strategy. To address this skew in ad delivery, we implemented the budget split design to allocate budget equally to sub-campaigns targeting male and female users. Our field test with the budget split demonstrated that **budget splitting reduced skew in the delivery ratio of the advertisers' campaigns, but also decreased cost effectiveness. However, targeting unknown users lessened the cost burden of budget splitting.**

Figure 2 demonstrates that with budget splitting, the share of spend and impressions attributed to male users shifts toward the advertisers' desired delivery skew of 0.5 in both Max Clicks (Figure 2a) and Max Conversions (Figure 2b) campaigns. In Figure 2 we show for each of four advertising engagement metrics (y-axis) the skew (x-axis) for three types of campaign strategies, which differ in their targeting granularity. The "All Users" condition reflects the original campaign, in which there was no gender-specific targeting applied. The "Single Gender + Unknown" condition contains the set of sub-campaigns created for the budget split field test which target unknown users along with either male or female users. The "Single Gender" condition contains the set of sub-campaigns created for the budget split field test which target male or female users alone (without unknown users). These are targeting strategies differ in granularity, with "All Users" as the least granular, "Single Gender + Unknown" as moderately granular, and "Single Gender" as the most granular. Skew for the budget split field test can be measured by aggregating engagement metrics from the "Single Gender + Unknown" and "Single Gender" conditions. Initially, the share of spend attributed to male users in the original Max Clicks campaign ("All Users") was 0.563 (99% CI=[0.55, 0.576]), and the share in the Max Conversions campaign was 0.462 (99% CI=[0.45, 0.474]). For the budget split aggregate results (aggregating "Single Gender + Unknown" and "Single Gender"), the target ratio of 0.5 was captured in the 99% CI of both Max Clicks (99% CI=[0.490, 0.534]) and Max Conversions (99% CI=[0.476, 0.517]) campaigns. We also observe the target ratio was included in the 99% confidence intervals for both the "Single Gender" and "Single Gender + Unknown" sub-campaigns separately. Our finding confirms prior literature that budget splitting is a functional approach to increasing advertiser control over the distribution of ad delivery, as the budget split field test did recover the ideal spend ratio.

While the budget split enforced parity in spend, we observe persistent skew in ad delivery. The budget split shifted impression skew toward the desired ratio, but in the Max Clicks campaign, impressions continued to skew toward male users (99% CI=[0.523, 0.541]). The initial skew in impressions was 0.55 (99% CI=[0.548, 0.554]). Impressions in the "Single Gender" campaigns, for both Max Clicks and Max Conversions, skewed toward male users. This occurred despite the initial skew toward female users in the Max Conversions campaign. This persistent gap is driven in part by differences in cost effectiveness: although we enforced parity in spend, male and female users have different costs to target. The cost effectiveness of reaching a demographic group is determined by a suite of factors, including predicted relevance, user's baseline engagement, and competitive spillover from other advertisers. Previous research has found that cost effectiveness varies by demographic group, with female users often costing more than male users [1, 5]. Consistent with prior literature, in the case of "Single Gender" targeting female users had lower cost effectiveness than male users. However, when users were targeted in the "Single Gender + Unknown" condition, we did not observe a difference in cost effectiveness by gender.

Figure 3a and Figure 3b demonstrate that cost increases as targeting granularity increases, and that the gap between male and female user cost is greatest under "Single Gender" targeting. In Figure 3, the height of each bar reflects the average weekly cost per 1,000 impressions (CPM), which is calculated as spend attributed to a given subgroup (i.e. male or female users) divided by the number of impressions attributed to that subgroup. This value is then multiplied by 1,000. The x-axis contains the three types of campaign targeting conditions: "All Users", "Single Gender + Unknown" and "Single Gender". Our case study demonstrates that sub-campaigns that target single-gender users along with unknown users provide a

middle ground for budget-splitting in which advertisers can reduce skew, relative to the "All Users" condition, without incurring the costs of the "Single Gender" condition. Furthermore, our findings demonstrate the practical necessity of continuously updating the budget split across sub-campaigns, as cost effectiveness may change under budget splitting. Otherwise, when the cost effectiveness of one subgroup changes relative to another, iterative budget updates are necessary to avoid exacerbating skew.

Conclusion Our research addresses fairness in advertising within a high-stakes context where budget splitting can directly improve access to business opportunities for underrepresented entrepreneurs. However, the budget split approach incurs substantial costs for advertisers, thus reducing the total number of users who will be reached by the advertising campaign. While prior literature on the ideal audit of ad delivery has limited study to groups of users with known demographic labels, we consider a real-world case in which excluding unknown users is not a viable option for the advertiser, as it would diminish advertising utility and exclude a large swath of the market. Instead we develop a non-ideal approach to budget splitting that introduces uncertainty by including unknown users, but also lessens the trade-offs for the advertiser. In particular, we found that while budget splitting tests create significant cost burdens for advertisers, targeting unknown users shows promise as a middle-road option that strikes a balance between advertiser public interest and campaign utility. By targeting unknown users, advertisers can alleviate the cost burden of budget splitting and reduce the disparity in cost effectiveness between male and female users.

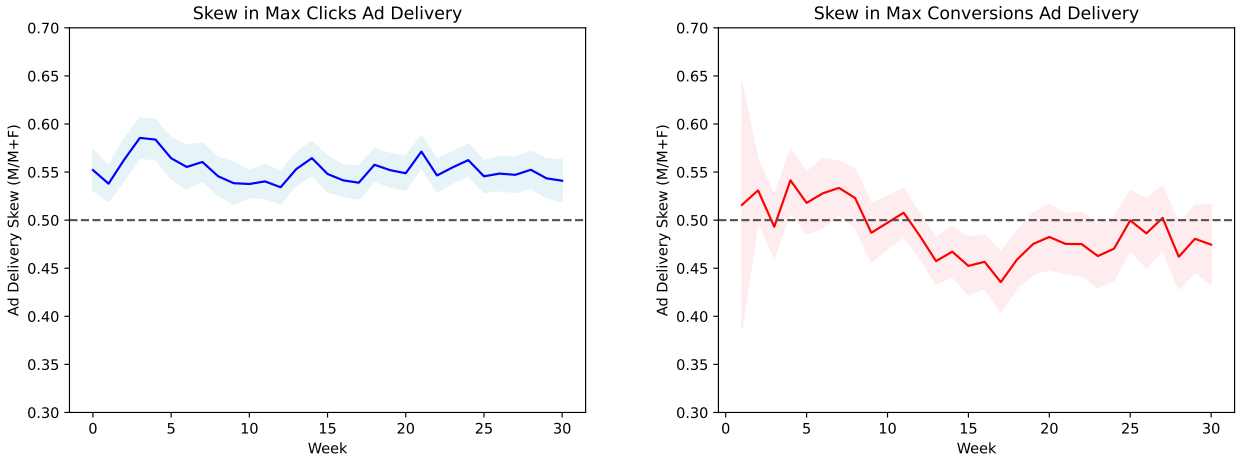

Figure 1: The original Max Clicks campaign, which targeted all users (male, female, and unknown) consistently delivered a greater share of ads to male users than female users over the 31 weeks for which the campaign was active. The original Max Conversions campaign did not have consistent skew over the same period.

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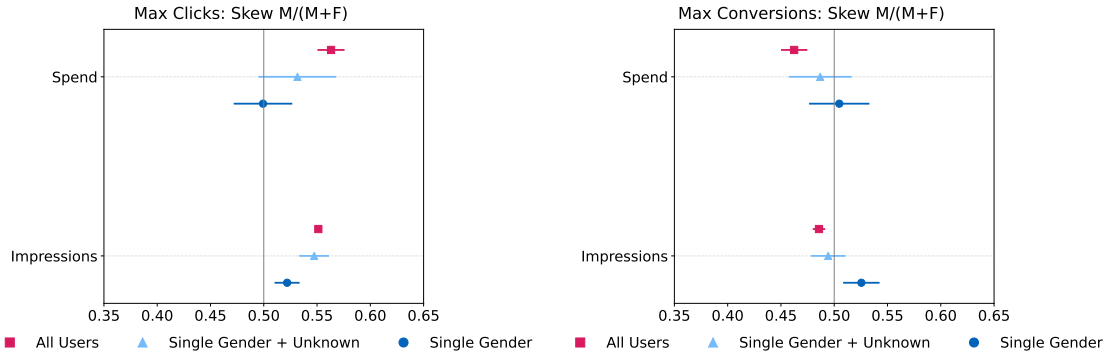
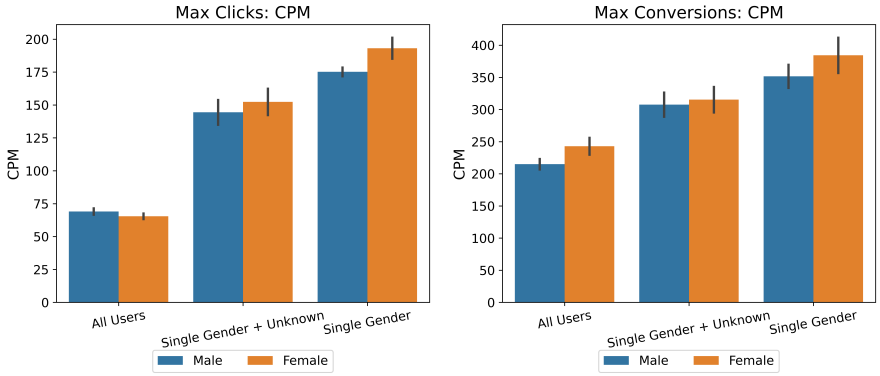


(a) **(b)**

Figure 2: Budget splitting reduced skew in the delivery ratio of Max Clicks and Max Conversions campaigns (a,b). Budget splitting including "unknown" demographics serves as a useful middle ground between no budget splitting and budget splitting with only known genders. Figures a, b show the share of male users for each level of targeting ("All Users", "Single Gender + Unknown" and "Single Gender"). 99% CI with an Agresti and Coull correction are shown.



(a) **(b)**

Figure 3: Budget splitting increased cost per 1,000 impressions (CPM). Budget splitting including "unknown" demographics serve as a useful middle ground between no budget splitting and budget splitting with only known genders. Figure shows average weekly CPM for campaigns with each targeting specification: "All Users", "Single Gender + Unknown" and "Single Gender". Standard error bars are shown.

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